

COMPUTER PROGRAMMING

Electrical Engineering Department — UET Nowshera

LAB 1

Lab Title: Installation of DEV-C++ IDE and Execution of First C++ Program

1. OBJECTIVE

The purpose of this lab is to gain hands-on experience with installing the DEV-C++ Integrated Development Environment (IDE) and running a fundamental C++ program. Students are introduced to the process of writing, compiling, and executing a basic C++ program within the DEV-C++ environment.

2. THEORY

C++ is a powerful, general-purpose programming language extensively used in the development of software and engineering solutions. In order to write, compile, and execute C++ programs, a suitable programming environment called an Integrated Development Environment (IDE) is essential.

DEV-C++ is an easy-to-use IDE designed for beginners, offering a unified environment to write, compile, and run programs without switching between tools. It includes a source-code editor, an integrated compiler, and a dedicated output window. In C++, program execution always begins from the `main()` function. The `cout` statement is responsible for displaying output on the console screen.

3. SOFTWARE USED

- DEV-C++ IDE
- C++ Compiler (bundled with DEV-C++)

4. PROCEDURE

Step 1 — Install DEV-C++

- Visit the official website and download the DEV-C++ installation package.
- Execute the installer and carefully follow each step shown on the screen.
- After installation finishes, open the DEV-C++ application.

Step 2 — Create a New Source File

- Navigate to File → New → Source File to open a new, empty editor window.
- Use File → Save As to save the file, naming it lab1_hello.cpp.

Step 3 — Write the C++ Program

Type the code below into the editor window. The inline comments (beginning with `//`) describe what each line does:

```
// Include the standard input/output stream library (needed for cout)
#include <iostream>

// Allow use of standard library names without the 'std::' prefix
using namespace std;

// main() - the entry point of every C++ program
int main()
{
    // cout prints text to the console window
    cout << "Hello World!";

    // return 0 signals that the program ended successfully
    return 0;
}
```

Step 4 — Compile and Run

- Select Execute → Compile to build the source code and verify there are no errors.
- Select Execute → Run to launch the compiled program.
- The console output window should show: Hello World!

5. GITHUB TASK

The following version-control tasks were carried out on GitHub as part of this lab exercise:

- Registered a new GitHub account at github.com.
- Set up a new repository under the name Cpp-Lab-Tasks.
- Added the lab1_hello.cpp source file to the repository.
- Submitted the repository link to the instructor for evaluation and record purposes.

Repository reference: CP26_Lab_01

6. CONCLUSION

This lab provided a practical introduction to C++ programming using the DEV-C++ IDE. The program was successfully compiled and executed, producing the expected output 'Hello World!' on the console. Through this exercise, key C++ constructs were understood: the role of `#include <iostream>` for enabling input/output operations, using namespace `std` for simplified syntax, the `main()` function as the program entry point, `cout` for displaying output, and `return 0` as a standard termination signal. In addition, the GitHub workflow for lab submission and source-code management was practiced.